TIANTONG SUN

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EDUCATION

George Mason University, Fairfax, VA **2025–2027 (Expected)**
M.S. in Computer Science (in progress)

Huazhong University of Science and Technology (HUST), Wuhan, China **2021–2025**
B.S. in Mathematics and Applied Mathematics; **Qiangji Program (National Strong Foundation Plan)**; **Project 985 University**
Rank: **1** — GPA: **4.41/5.00**

RESEARCH INTERESTS

Trustworthy AI; federated learning; LLM post-training; agent harness optimization; personalized agent.

RESEARCH PAPERS AND CONTRIBUTIONS (Research Intern, Xiaojin Zhang's lab, HUST; Sep. 2024–Present)

FUSE: Fairness-aware Update Screening against Byzantine Exploitation in q-Fair Federated Learning
First author — *Supervised by Wei Chen, Xiaojin Zhang, Qiang Yang.* [arXiv] *ICLR 2027 (under review)*
My contribution: Formulated the research problem; proposed the core idea and FUSE algorithm; developed the theoretical proofs; designed and conducted all experiments.

FedPix: Learning What to Share with Less Leakage in Federated Intelligence
Co-first author — *Supervised by Wei Chen, Xiaojin Zhang.* [arXiv] *AAAI 2027 (under review)*
My contribution: Contributed to formulating the core research problem; proposed the core algorithm and developed its theoretical proofs.

From Unlearnable Outcomes to Decodable Shortcut Codes
Second author — *Supervised by Wei Chen, Xiaojin Zhang.* [OpenReview] *ICLR 2027 (under review)*
My contribution: Analyzed the problem, formulated the optimization problem, and proposed the algorithm.

OTHER PROJECTS

Personalized Federated Harness *Sep. 2026–Present — Ongoing*
Investigating personalized agent harnesses in federated settings, motivated by diverse user needs, task requirements, and underlying LLM capabilities.

AI-for-Math Summer Program, Peking University *Jun.–Sep. 2023*
Formalized **Chapters 1–2** of Neukirch's *Algebraic Number Theory* in **Lean**. [GitHub]

RESEARCH INTERNSHIP

Answer.AI *Jun.–Sep. 2023 — Research Intern, AI for Mathematics*
Implemented and evaluated lightweight algorithmic improvements for AI-based mathematical problem solving, with a focus on answer accuracy.

AWARDS AND SCHOLARSHIPS

National Scholarship 2022

Outstanding Graduate (HUST) 2025
School of Mathematics and Statistics

Outstanding Undergraduate Student 2022
Top 1%; selected for personalized research training

National Math Competition, 2nd Prize 2023

MCM Finalist Award (Top 1.7%) 2024

School Merit Student Scholarship (twice) 2022, 2023

LEADERSHIP AND ACTIVITIES

Executive President, Union of Outstanding Undergraduate Students *Sep. 2022–Present*

TECHNICAL SKILLS

Python, PyTorch, Huggingface, PEFT / LoRA, Lean, Linux, HPC / Slurm, MATLAB.